IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

DIGITAL SIGNAL PROCESSING AND DIGITAL FILTERS

Wednesday, 1 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer Question 1 and any TWO other questions.

Question 1 is worth 40% of the marks and other questions are worth 30%

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : A. Bhandari
Second Marker(s) : T. Stathaki

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Department of Electronic and Electrical Engineering

Examinations 2024

Digital Signal Processing and Digital Filters

MSc and EEE/EIE PART IV

May, 2024 • Exam Duration: Three Hours

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Answer Question 1 and any TWO other questions.
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Where unavoidable, make reasonable assumptions for your solutions.
Explain your assumptions in the answer script.

Markers: Ayush Bhandari and Tania Stathaki.

Datasheet:

Standard Sequences

- $\delta[n] = 1$ for $n = 0$ and 0 otherwise.
- $\delta_{\text{condition}}[n] = 1$ whenever "condition" is true and 0 otherwise.
- $u[n] = 1$ for $n \geq 0$ and 0 otherwise.

Geometric Progression

- $\sum_{n=0}^{r} \alpha^n z^{-n} = \frac{1-\alpha^{r+1} z^{-r-1}}{1-\alpha z^{-1}}$ provided that $\alpha z^{-1} \neq 1$.
- $\sum_{n=0}^{\infty} \alpha^n z^{-n} = \frac{1}{1-\alpha z^{-1}}$ provided that $\left|\alpha z^{-1}\right| < 1$.

Forward and Inverse Transforms

z:	$X(z) = \sum_{-\infty}^{\infty} x[n] z^{-n}$	$x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$
CTFT:	$X(j\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega) e^{j\Omega t} d\Omega$
DTFT:	$X(e^{j\omega}) = \sum_{-\infty}^{\infty} x[n] e^{-j\omega n}$	$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$
DFT:	$X[k] = \sum_{0}^{N-1} x[n] e^{-j2\pi \frac{kn}{N}}$	$x[n] = \frac{1}{N} \sum_{0}^{N-1} X[k] e^{j2\pi \frac{kn}{N}}$
DCT:	$X[k] = \sum_{n=0}^{N-1} x[n] \cos \frac{2\pi(2n+1)k}{4N}$	$x[n] = \frac{X[0]}{N} + \frac{2}{N} \sum_{n=1}^{N-1} X[k] \cos \frac{2\pi(2n+1)k}{4N}$
MDCT:	$X[k] = \sum_{n=0}^{2N-1} x[n] \cos \frac{2\pi(2n+1+N)(2k+1)}{8N}$	$y[n] = \frac{1}{N} \sum_{0}^{N-1} X[k] \cos \frac{2\pi(2n+1+N)(2k+1)}{8N}$

Convolution

DTFT:	$v[n] = x[n] * y[n] \triangleq \sum_{r=-\infty}^{\infty} x[r] y[n-r]$	$\Leftrightarrow$	$V\left(e^{j\omega}\right) = X\left(e^{j\omega}\right) Y\left(e^{j\omega}\right)$
	$v[n] = x[n] y[n]$	$\Leftrightarrow$	$V\left(e^{j\omega}\right) = \frac{1}{2\pi} X\left(e^{j\omega}\right) \circledast Y\left(e^{j\omega}\right) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X\left(e^{j\theta}\right) Y\left(e^{j(\omega-\theta)}\right) d\theta$
DFT:	$v[n] = x[n] \circledast_N y[n] \triangleq \sum_{r=0}^{N-1} x[r] y[(n-r)_{\mod N}]$	$\Leftrightarrow$	$V[k] = X[k] Y[k]$
	$v[n] = x[n] y[n]$	$\Leftrightarrow$	$V[k] = \frac{1}{N} X[k] \circledast_N Y[k] \triangleq \frac{1}{N} \sum_{r=0}^{N-1} X[r] Y[(k-r)_{\mod N}]$

Group Delay

The group delay of a filter, $H(z)$, is $\tau_H(e^{j\omega}) = -\frac{d\angle H(e^{j\omega})}{d\omega} = \Re\left(\frac{-z}{H(z)} \frac{dH(z)}{dz}\right)\Big|_{z=e^{j\omega}} = \Re\left(\frac{\mathscr{F}(nh[n])}{\mathscr{F}(h[n])}\right)$ where $\mathscr{F}(\,)$ denotes the DTFT.

Order Estimation for FIR Filters

Three increasingly sophisticated formulae for estimating the minimum order of an FIR filter with unity gain passbands:

1. $M \approx \frac{a}{3.5\Delta\omega}$
2. $M \approx \frac{a-8}{2.2\Delta\omega}$
3. $M \approx \frac{a-1.2-20\log_{10} b}{4.6\Delta\omega}$

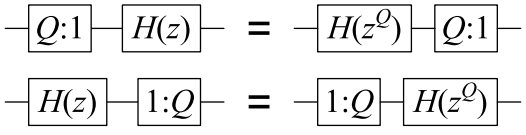
where a =stop band attenuation in dB, b = peak-to-peak passband ripple in dB and $\Delta\omega$ = width of smallest transition band in radians per sample.

z-plane Transformations

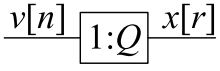
A lowpass filter, $H(z)$, with cutoff frequency ω_0 may be transformed into the filter $H(\hat{z})$ as follows:

Target $H(\hat{z})$	Substitute	Parameters
Lowpass $\hat{\omega} < \hat{\omega}_1$	$z^{-1} = \frac{\hat{z}^{-1}-\lambda}{1-\lambda\hat{z}^{-1}}$	$\lambda = \frac{\sin\left(\frac{\omega_0-\hat{\omega}_1}{2}\right)}{\sin\left(\frac{\omega_0+\hat{\omega}_1}{2}\right)}$
Highpass $\hat{\omega} > \hat{\omega}_1$	$z^{-1} = -\frac{\hat{z}^{-1}+\lambda}{1+\lambda\hat{z}^{-1}}$	$\lambda = \frac{\cos\left(\frac{\omega_0+\hat{\omega}_1}{2}\right)}{\cos\left(\frac{\omega_0-\hat{\omega}_1}{2}\right)}$
Bandpass $\hat{\omega}_1 < \hat{\omega} < \hat{\omega}_2$	$z^{-1} = -\frac{(\rho-1)-2\lambda\rho\hat{z}^{-1}+(\rho+1)\hat{z}^{-2}}{(\rho+1)-2\lambda\rho\hat{z}^{-1}+(\rho-1)\hat{z}^{-2}}$	$\lambda = \frac{\cos\left(\frac{\hat{\omega}_2+\hat{\omega}_1}{2}\right)}{\cos\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)}$, $\rho = \cot\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)\tan\left(\frac{\omega_0}{2}\right)$
Bandstop $\hat{\omega}_1 \not< \hat{\omega} \not< \hat{\omega}_2$	$z^{-1} = \frac{(1-\rho)-2\lambda\hat{z}^{-1}+(\rho+1)\hat{z}^{-2}}{(\rho+1)-2\lambda\hat{z}^{-1}+(1-\rho)\hat{z}^{-2}}$	$\lambda = \frac{\cos\left(\frac{\hat{\omega}_2+\hat{\omega}_1}{2}\right)}{\cos\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)}$, $\rho = \tan\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)\tan\left(\frac{\omega_0}{2}\right)$

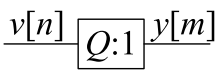
Noble Identities



Multirate Spectra

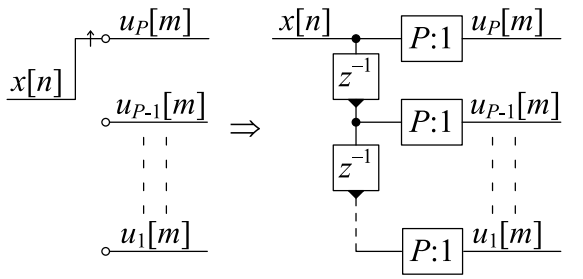
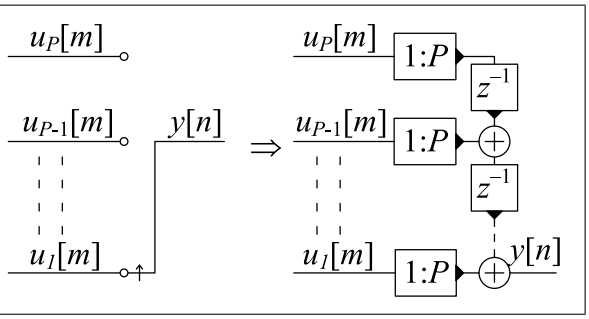
Upsample: 

$$x[r] = \begin{cases} v\left[\frac{r}{Q}\right] & \text{if } Q \mid r \\ 0 & \text{if } Q \nmid r \end{cases} \quad \Rightarrow \quad X(z) = V(z^Q)$$

Downsample: 

$$y[m] = v[Qm] \quad \Rightarrow \quad Y(z) = \frac{1}{Q}\sum_{k=0}^{Q-1} V\left(e^{\frac{-j2\pi k}{Q}} z^{\frac{1}{Q}}\right)$$

Multirate Commutators

Input Commutator	Output Commutator
	

Question 1 [40 Marks]

a) For a sequence $x[n], n = -N, \ldots, N$, the relationship between its average power $\mathcal{P}_x$ and energy $\mathcal{E}_x$ is given by

$$\mathcal{P}_x = \lim_{N\to\infty} \frac{\mathcal{E}_x}{2N+1}$$

where $\mathcal{E}_x = \sum_{|n|\leqslant N} |x\,[n]|^2$. Using this, compute the average power of the unit step sequence,

$$u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases}.$$

[2 Marks]

b) The input/output relation for a signal processing system is described by

$$\pi^{\pi} u\,[n] \longrightarrow \boxed{h\,[n]} \longrightarrow e^{\pi} \cos(\omega_0 n) u\,[n]$$

where $u[n]$ is the usual unit-step sequence.

Find the impulse response of this system. **[3 Marks]**

c) A certain filter h is defined as,

$$h[n] = \begin{cases} \lambda^n & n \geqslant 0 \\ 10\mu^{2n} & n < 0 \end{cases}.$$

Find the conditions on λ such that the filter is stable. **[3 Marks]**

d) Real-world signals **[2 Marks]**

i) vary continuously in time,
ii) are quantized in amplitude,
iii) are analog in amplitude,
iv) are discrete in time.

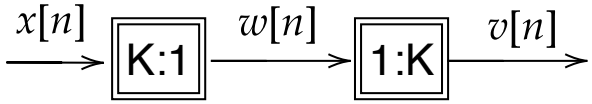
(Multiple answers are possible.)

e) A certain sequence is defined by

$$h\,[n] = \lambda^{-(n+\alpha)} u\,[n-\beta]$$

where α, β are positive integers and λ is a real-valued scalar.

i) Compute the discrete time Fourier transform (DTFT) of $h[n]$. **[3 Marks]**
ii) For what values of λ, $h[n]$ forms a stable filter? **[2 Marks]**



f) Figure above shows a downsampler followed by an upsampler.

i) Show that $V(z) = W(z^K)$. **[3 Marks]**

ii) Show that $c[n] = \frac{1}{K}\sum_{k=0}^{K-1} e^{\jmath\frac{2\pi kn}{K}}$ is equal to 1 if n is an integer multiple of K and equal to 0 otherwise. **[3 Marks]**

iii) Show that $W(z) = \frac{1}{K}\sum_{k=0}^{K-1} X(e^{-\jmath\frac{2\pi k}{K}} z^{\frac{1}{K}})$. **[4 Marks]**

g) Let us define a sequence by

$$x\left[n\right] = \left(r_1\right)^n u\left[n\right] - \left(r_2\right)^n u\left[-\left(n+1\right)\right] \tag{1}$$

where $r_1 = -1/3$ and $r_2 = 1/2$. On the z-plane, plot the poles and zeros together with the region-of-convergence.

[2 Marks]

In the above sequence in (1), what happens if we swap r_1 and r_2? **[3 Marks]**

h) For some linear-time-invariant system, the transfer function is given by

$$H\left(z\right) = \frac{\left(1+z^{-1}\right)^2}{\left(1-\frac{1}{2}z^{-1}\right)\left(1+\frac{3}{4}z^{-1}\right)}.$$

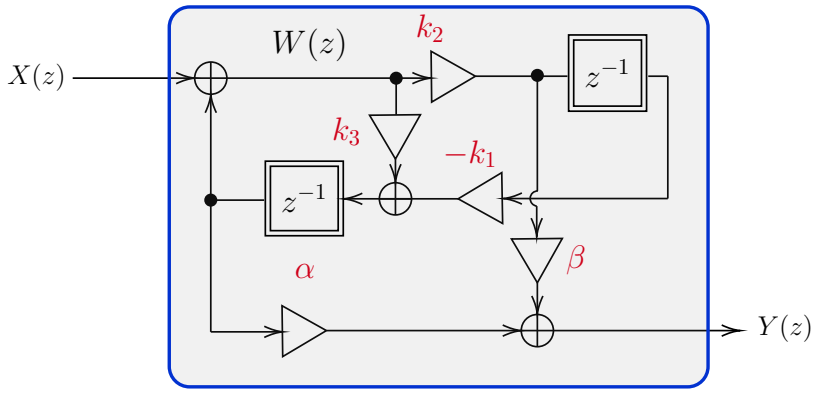
Suppose $x\left[n\right]$ is the input to the system and $y\left[n\right]$ is the output. Derive the difference equation that is satisfied by $x\left[n\right]$ and $y\left[n\right]$. **[4 Marks]**

i) Identify parameters p and q in the filter, $h\left[n\right] = pq^n u\left[n\right]$ given that $h\left[1\right] = 1$ and the group delay at $\omega = \pi$ is $-1/3$. **[2 Marks]**

j) Let $h\left[k\right]$ be a filter defined by $h\left[k\right] = \frac{(-1)^k}{k}, k > 0$ and $h\left[k\right] = 0, k \leq 0$. Is this system BIBO-stable? If not, demonstrate it for a particular input sequence. [Hint: the harmonic series is divergent $1 + 1/2 + 1/3 + ... = \infty$]. **[4 Marks]**

Question 2 [30 Marks]

a) Let $H(z) = \frac{Y(z)}{X(z)}$ be a system represented in the diagram below



i) Derive the expression of $W(z)$ in terms of $X(z)$. **[3 Marks]**

ii) Derive the expression of the transfer function $H(z)$. **[4 Marks]**

iii) What are the conditions that ensure that $H(z)$ is BIBO-stable? **[3 Mark]**

b) *i*) Let γ be a complex-valued constant, compute the z-transform of a causal sequence $x[n] = e^{\gamma^2 n}$ and specify its region of convergence. **[2 Marks]**

ii) Suppose that $G_1(z) = 1 - az^{-1}$ and $G_2(z) = 1 - bz^{-1}$ where the constants a and b may be complex. If $ab^* = 1$, show that $|G_1(e^{\jmath\omega})| = p|G_2(e^{\jmath\omega})|$ for all ω and determine the constant p. **[3 Marks]**

iii) Let $H_1(z) = 4z^{-2} - 9z^{-1} + 2$. Find the expression of $H_2(z)$ such that $|H_1(e^{\jmath\omega})| = |H_2(e^{\jmath\omega})|$ for all ω and that all the zeros of $H_2(z)$ lie inside the unit circle. **[3 Marks]**

iv) The impulse response, $h[n]$, of a linear time-invariant system satisfies $\sum_{n=-\infty}^{\infty} |h[n]| = \alpha < \infty$. Show that the system is BIBO-stable and also that $H(z)$ converges for $|z| = 1$. **[3 Marks]**

c) A continuous-time filter has the transfer function $H(s) = \frac{1}{s^2+\sqrt{2}s+1}$.

i) Show that the gain of the filter, $|H(\jmath\Omega)|$, equals -3dB at an angular frequency $\Omega = 1$. **[1 Marks]**

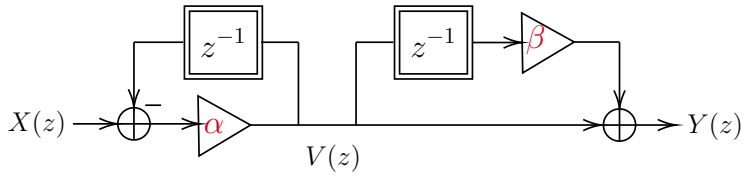
ii) Using the bilinear mapping, $s = 2f_s\frac{z-1}{z+1}$, determine the transfer function, $G(z)$, of the corresponding discrete-time filter. **[3 Marks]**

iii) Determine the sample frequency, f_s, so that the discrete-time filter has a gain of -3dB at an angular frequency $\omega = 1$. **[2 Marks]**

iv) For the sample frequency found in (iii) above, give the coefficients of $G(z)$ to 3 decimal places when the coefficients are normalized so that the leading coefficient of the denominator is unity. **[3 Marks]**

Question 3 [30/30 Marks]

a) Let $H(z) = \frac{Y(z)}{X(z)}$ be represented by the diagram below.



i) Derive the expression of $V(z)$ **[3 Marks]**

ii) Derive the expression of the transfer function $H(z)$. **[3 Marks]**

iii) Give conditions for which $H(z)$ realizes a first order all-pass transfer. **[2 Marks]**

iv) What are the conditions that ensure that $H(z)$ is BIBO-stable? **[2 Mark]**

b) Let $y[k]$ be the output of a system $\mathcal{H}$ satisfying $\mathcal{H}x[k] = y[k] = x[k+1] - x[k]$. The system is also known as the first order finite difference.

i) Prove that the system is linear and time-invariant. **[3 Marks]**

ii) Compute the system transfer function $H(z)$. Compute analytically its magnitude and phase response functions. **[3 Marks]**

iii) Let $\mathcal{H}_N$ be defined recursively for $N \in \mathbb{Z}, N \geq 1$ as $\mathcal{H}_{N+1} = \mathcal{H}_N * \mathcal{H}$, and $\mathcal{H}_1 = \mathcal{H}$. Here $\mathcal{H}_N$ is known as the N-th order finite difference filter. Compute the transfer function $H_N(z)$ of $\mathcal{H}_N$. Furthermore, derive the magnitude and phase response of $H_N(z)$. **[3 Marks]**

iv) Find all $\omega \in \mathbb{R}^+$ such that $\lim_{N\to\infty} H_N\left(e^{j\omega}\right) = 0$. **[2 Marks]**

c) Let $H(z) = H_1(n) + z^{-N} \cdot H_2(z)$ where $H_1(z)$ and $H_2(z)$ are two filters satisfying

$$H_1(z) = 1 - 2z^{-1} + \alpha z^{-2} + 2z^{-3}, \tag{2}$$

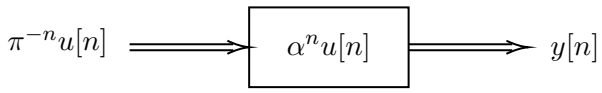
$$H_2(z) = -1 + 5z^{-1} + z^{-2} - z^{-3}. \tag{3}$$

i) Compute α and N such that $h[n]$ has a linear phase response. **[4 Marks]**

ii) Derive analytically the phase and gain response of $h[n]$. **[5 Marks]**

Question 4 [30 Marks]

a) A signal processing system is described by



i) Under suitable conditions on α, compute the DTFT of the output. **[3 Marks]**

ii) Explain the choice of suitable conditions on α in the above question. **[2 Marks]**

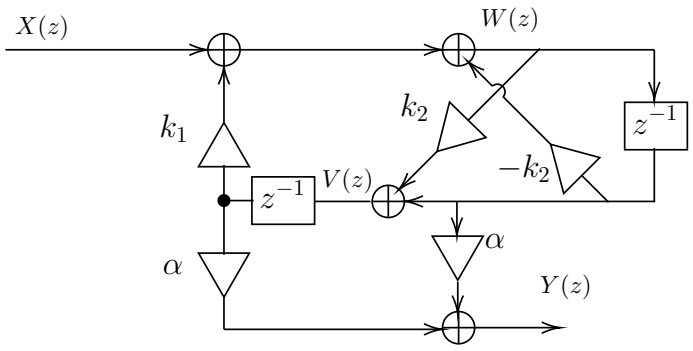
iii) Write the explicit form of $y[n]$ in terms of α. **[3 Marks]**

iv) Find the value of α given that,

$$y\left[1\right] = \frac{\pi e}{\pi + e}$$

where e is the Euler's number. **[2 Marks]**

b) Let $H\left(z\right) = \frac{Y(z)}{X(z)}$ be a system represented in the diagram below



i) Derive the expression of $V\left(z\right)$ and $W\left(z\right)$ **[4 Marks]**

ii) Derive the expression of the transfer function $H\left(z\right)$. **[4 Marks]**

iii) What are the conditions that ensure that $H\left(z\right)$ is BIBO-stable? **[2 Marks]**

c) Let $H\left(z\right) = H_1\left(z\right) + z^{-1}H_2\left(z\right)$ where $H_1\left(z\right)$ and $H_2\left(z\right)$ are two filters satisfying

$$H_1\left(z\right) = \beta/2 + z^{-1} - z^{-2} - z^{-4}, \tag{4}$$

$$H_2\left(z\right) = 2 + (\gamma - 3)z^{-1} - 4z^{-3} + 3z^{-4} - 2z^{-5}. \tag{5}$$

i) Compute β and γ such that $h\left[n\right]$ has a linear phase response. **[4 Marks]**

ii) Derive analytically the phase and gain response of $h\left[n\right]$. **[4 Marks]**

d) Let $y_N\left[k\right]$ be the output of a system with impulse response $h_N\left[k\right]$ satisfying

$$y_N\left[k\right] = \sum_{n=1}^{N} \frac{x\left[k-n\right]}{n(n+1)}.$$

Prove that the system is linear and time-invariant. **[2 Marks]**